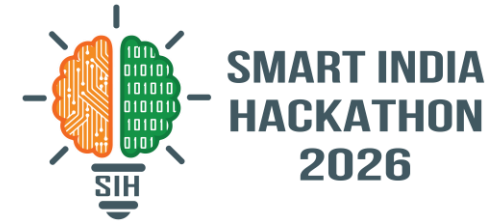


SMART INDIA HACKATHON 2026



TITLE PAGE

Problem Statement ID – 26032

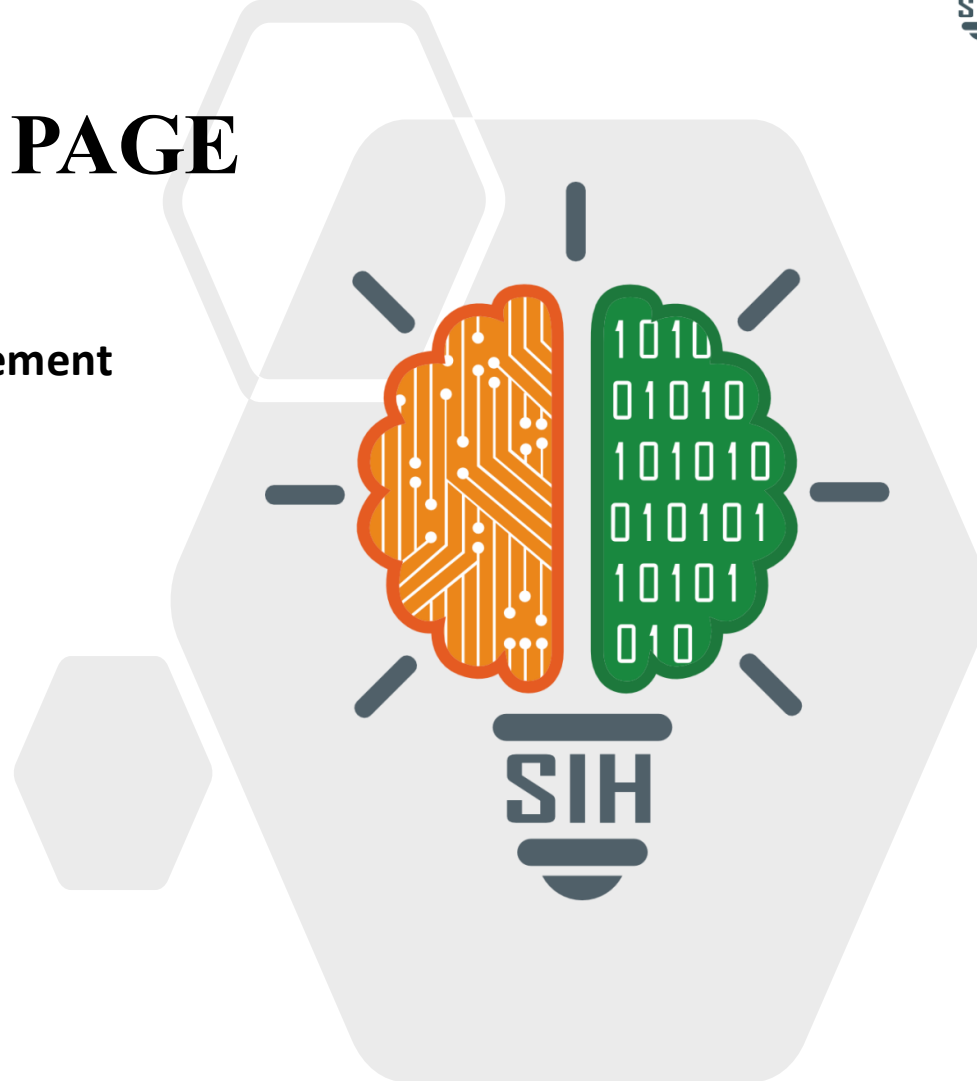
**Problem Statement Title – Smart Procurement Management
for Govt. Mandis**

Theme – Smart Automation

PS Category – Software

Team ID – [As on SIH Portal]

Team Name – KisanSetu AI





❖ Proposed Solution

- KisanSetu AI — an end-to-end smart procurement & queue management platform for Indian APMC Mandis
- Farmers book time-slots digitally from home → receive a unique digital queue token (e.g. A014)
- AI engine scores each slot for congestion (0-100) and recommends the lowest-wait option
- Real-time WebSocket queue board, Desktop Tray Notifier, & phone SMS — zero manual refresh
- Officer digitally grades crop (Grade A / Standard / Moisture %) → instant digital J-Form receipt
- Direct Benefit Transfer (DBT) tracking: PENDING → PROCESSING → COMPLETED with txn ref
- Government admin sees live national KPI dashboard — congestion heatmap, volume, payment rates

Innovation: 3-Layer AI Congestion Engine | Desktop Tray App | Multi-Channel SMS (ADB/Cloud) | Zero Paper

Live App: <https://kisansetu-sih.vercel.app>

- **Tech Stack**

Frontend: Next.js 16 (TypeScript) · Tailwind CSS · App Router · WebSocket client

Backend: FastAPI (Python 3.13) · SQLAlchemy 2.0 async · Pydantic v2 · JWT auth

Database: SQLite (dev) / PostgreSQL (prod) · Supabase CLI · Zero-downtime store

Infra: Docker Compose · GitHub Actions CI/CD · Vercel · System Tray Notifier daemon

APIs: REST (12 routers) + 2 WebSocket channels + SMS Gateway (ADB / MSG91)

Methodology / Flow

Farmer → Books Slot via AI Recommendation → Receives Digital Token

Officer → Calls Token via Queue Desk / Tray Notifier → Digital Grading → J-Form

System → Instant WebSocket push + Phone SMS alert → DBT Payment PENDING

Admin → Live Heatmap updates → Monitors Mandi KPIs & Congestion in real-time

Prototype: Fully functional — 15 Mandis, 19 crops, 1200+ slots, Tray App · 13 tests passing 

- **Feasibility**

- ✓ Tech stack is proven, open-source, and free-tier deployable (Vercel + Supabase)
- ✓ One-click launcher (start.bat) boots Backend, Frontend, and Tray Notifier concurrently
- ✓ GitHub Actions auto-refreshes 14-day slot schedules weekly — zero cloud shutdown
- ✓ Scales to 7,000+ APMC Mandis via PostgreSQL read-replicas and distributed caching

Challenges & Mitigations

Challenge: Low smartphone penetration in rural farming communities

→ Solution: Multi-channel SMS engine (ADB USB bridge + MSG91 DLT gateway) reaches any 2G phone

Challenge: Internet connectivity / server downtime at remote Mandis

→ Solution: Resilient dual-layer store (local SQLite / Next.js store) ensures 100% offline uptime

Challenge: Farmer adoption & digital literacy

- **Impact & Benefits**

Social Impact

- Average mandi wait time reduced by 23.3% (simulation: 241 min → 185 min)
- Eliminates 12–48 hour tractor roadblocks at peak Rabi / Kharif harvest seasons
- Zero post-harvest produce spoilage from multi-day open-air queue bottlenecks

Economic Impact

- Tamper-proof digital J-Form receipts protect farmers against unauthorised deductions
- Direct Benefit Transfer (DBT) tracking prevents middleman & commission-agent leakage
- Target: 7,000+ APMC Mandis across India → millions of farmers directly benefited

Operational / Government Impact

- Real-time national KPI dashboard gives DoCA & Ministry officers live actionable data
- Paperless procurement audit trail — instant verification, weighbridge to bank credit

- **Research & References**

1. SIH 2026 Problem Statement 26032 — Ministry of Consumer Affairs, Food & Public Distribution
<https://www.sih.gov.in>
2. APMC Mandi Procurement & MSP Policy — Food Corporation of India (FCI)
<https://fci.gov.in>
3. Congestion Reduction Simulation — KisanSetu AI Internal Analysis (2026)
docs/CONGESTION_REDUCTION_ANALYSIS.md · 23.3% avg wait-time reduction validated
4. CACP Minimum Support Prices (2024–2026) — Commission for Agricultural Costs & Prices
5. Tech Stack: FastAPI 0.115 · Next.js 16 · Supabase PostgreSQL · WebSocket RFC 6455

Live App: <https://kisansetu-sih.vercel.app>

GitHub: <https://github.com/YTxFSGAMERz/KisanSetu-AI>

@SIH Idea submission- Template